

CHAPTER 15: POWER GENERATION/ DEMONSTRATE UNDERSTANDING OF POWER GENERATIONS

15.1. Introduction of the Unit of Learning / Unit of Competency

Power generation entails the use of energy resources to produce electricity. This competency in power generation is applied in electricity generating sectors. This unit of competency covers sources of energy, power plants and their linkages in electrical engineering. The essential resources required in this unit include video tapes of power generation, models of power station, electrical charts of power generating stations, and actual visits to the power generating stations. Upon the completion of this course the trainee is expected to establish the design of a power station, operate and maintain a power station while observing the safety procedures and standards applicable in power generating plants. This course prepares the trainee for a gainful career in power generating sectors.

15.2. Performance Standard

demonstrate understanding of power generating station layout as per the source of energy, expected plant output and the safety involved as per IEEE regulation, demonstrate understanding in the operation of power generating stations constituents as determined by the type of the generating plant and their standard operating procedure Finally, the trainee will demonstrate understanding in the operating sequence of generating station as determined by the energy flow, type of generating stations, and components used in the generating station.

15.3. Learning Outcome

15.3.1. List of Learning Outcomes

- a) Identified types of generating stations
- b) Demonstrated understanding of power generating station layout
- c) Demonstrated understanding in the operation of power generating stations
- d) Demonstrated understanding in the operating sequence of generating stations.

15.3.2 Learning Outcome No. 1. Identified types of generating stations

15.3.2.1 Learning Activities

Learning Outcome No. 1. Identified types of generating stations		
Learning Activities		Special Instructions
i)	Identifying types of generating stations as per their sources of energy form visiting websites ,selecting plants	-visit appropriate website -use appropriate video tapes
ii)	Identifying the appropriate generating station for a given location as in the line with their source of energy. Through visiting websites, using video tapes and visiting potential sites and	-Consult trainers for visits outside the training workshops

15.3.2.2 Information Sheet No. 15/ LO1

Definition of terms

Generation: It is the process of power production by use of an alternator.

Alternator: A device which produces alternating electrical current.

Current: It's the flow of charges in the circuit.

Power station: It is a unit where the alternator, control and protection devices are installed for the purpose of electricity generation.

Transformer: A device for stepping up and stepping down the voltage.

Sources of energy

- water
- wind
- Geothermal
- Solar
- Nuclear
- Diesel
- Waves
- Biomass
- Tides

Factors consider in generating plant setup;

- Availability of the source of power
- The capital available
- The security of the area
- The available technology
- Government policies
- The legal aspect
- The population of the area

Types of generating stations as per their sources of energy

1. Geothermal power station
2. Hydroelectric power station
3. Wind energy power station
4. Solar energy power station
5. Nuclear power stations
6. Diesel engine power station
7. Thermal power station
8. Waves power station
9. Waste water and Bio gas power station
10. Tidal power station

Generating station location as in the line with their source of energy.

- a) Geothermal power station e.g. Olkaria I, Olkaria II, Olkaria III and Olkaria IV located at Olkaria-Naivasha-Kenya where geothermal energy is harvested.
- b) Hydroelectric power station e.g. -Kindaruma, masinga, Kiambere, Gitaru power station and Kamburu power station located along Tana river
 - Wanjii power station along Mathioya River
 - Sondu Miriu power station along Sondu River
- c) Wind energy power station-e.g. –Lake Turkana Wind power project (LTWP) at Loiyangalani District, in Marsabit county-Kenya.
- d) Solar energy power station e.g. Garissa power station(55MW) in Garissa County
- e) Waves power station e.g. 2.5 MW wave farm Aguçadoura of the coast of Portugal(IRENA Ocean Energy Technology Brief 4June 2014)

Tidal power station e.g. Sihwa Lake tidal power station (2 August 2016 by Yun Ha Kim)

15.3.2.3. Self-Assessment

1. Which of the following is the correct equation for the electrical power generated by the hydroelectric power plant?
 - a) $75 \times 0.736 \text{ wQH}\eta$ Watt
 - b) $(7.5/0.736) \times \text{wQH}\eta$ Watt
 - c) $0.845 \times \text{wQH}\eta$ Watt
 - d) $9.81 \times \text{wQH}\eta$
2. Which of the following is not a requirement for site selection of hydroelectric power plant?
 - a) Availability of water
 - b) Large catchment area
 - c) Rocky land
 - d) Sedimentation
3. The amount of electrical energy that can be generated by a hydroelectric power plant depends upon _____
 - a) Head of water

- b) Quantity of water
 - c) Specific weight of water
 - d) Efficiency of Alternator
4. Potential energy of water is used to drive the turbine.
 - a) True
 - b) False
 5. Hydroelectric power plant is _____
 - a) Non-renewable source of energy
 - b) Conventional source of energy
 - c) Non-conventional source of energy
 - d) Continuous source of energy
 6. Hydroelectric power plant is generally located near load centre.
 - a) True
 - b) False
 7. Hydroelectric power plant is mainly located in _____
 - a) Flat areas
 - b) Deserts
 - c) Hilly areas
 - d) Deltas
 8. Which statement about hydroelectric power plant is wrong?
 - a) Efficiency of hydroelectric power plant does not reduce with age
 - b) Its construction cost is very high and takes a long time for erection.
 - c) It is very neat and clean plant because no smoke or ash is produced.
 - d) Meeting rapidly changing load demands is not possible in hydroelectric power plant.
 9. Which of the following is not an advantage of hydroelectric power plant?
 - a) no fuel requirement
 - b) low running cost
 - c) continuous power source
 - d) no standby losses
 10. Which of the following statement is true about hydroelectric power plant?
 - a) Hydroelectric power plants are multipurpose.
 - b) Due to non-uniform flow of water frequency control in such plants is very difficult.
 - c) Hydroelectric power plant has high running cost
 - d) Water is used as fuel in hydroelectric power plant
 11. State FIVE types of generating stations
 12. Give five sources of electrical energy
 13. In each case, give the type of generating station and the resource used for power generation
 14. Explain the factors to consider when selecting a site for a power station
 15. Explain factors consider in generating plant setup
 16. Describe types of generating stations as per their sources of energy

Practical exercises

1. Identifying types of generating stations as per their sources of energy form visiting websites ,selecting plants
2. Identifying the appropriate generating station for a given location as in the line with their source of energy. Through visiting websites, using video tapes and visiting potential sites and

15.3.2.4. Tools, Equipment, Supplies and Materials for the specific learning outcome

- Models of generating station
- Charts
- Actual generating stations
- Video tapes
- Actual of power plant
- OSHA standards
- IEE regulations

15.3.2.5. References

1. Principles of Power systems by V.K Mehta.
2. Sihwa Lake tidal power station (2 August 2016 by yun ha Kim)

15.3.3. Learning Outcome No. 2. Power generating station layout

15.3.3.1. Learning Activities

Learning Outcome No. 2. Power generating station layout	
Learning Activities	Special Instructions
The trainee to observe a video from the following links or any other appropriate link for power generation plant and hence : a) Observing a model power station, charts, video tape and drawing the layouts and generating station as per the source of energy b) Determining the plant output c) Observing model sites, charts, video tapes, visiting potential site for generating station and determining the appropriate plant layout as per the safety involved	-visit a power plant -visit the various links -Observe model power stations -Observe video tapes -Visit different websites -Hold consultations and discussions

15.3.3.2 Information Sheet No 15/LO2.

Establishing layout of generating station as per the source of energy

Types of Geothermal power Station

- Dry Steam Power Plant
- Flash Steam Power Plant
- Binary Cycle power plant
- Hydro Power Station
- A Thermal Power Plant
- Wind Power Station
- Nuclear Power Station
- Diesel Engine Power Station
- Solar Power Station
- Tidal Power Station

Determining the plant output for a given layout.

Wind power plant

Equation for wind power is given below;

$$P = \frac{\rho A V^3}{2} \quad \text{where } \rho - \text{Density of air}$$

The more the dense the air, the more energy is received by the turbine.

A-Swept area of the turbine

The large the swept area, the more power the turbine can capture from the wind.

V-Wind speed

The amount of energy in the wind varies with the cube of the wind speed.

Hydro power station

$$P = m \times g \times H_{\text{net}} \times \eta$$

Where:

P- Power measured in Watts (W).

m- Mass flow rate in kg/s (numerically the same as the flow rate in litres/second because 1 litre of water weighs 1 kg)

g - The gravitational constant, which is 9.81 m/s^2

H_{net} - the net head. This is the gross head physically measured at the site, less any head losses. To keep things simple head losses can be assumed to be 10%, so

$$H_{\text{net}} = H_{\text{gross}} \times 0.9$$

H - The product of all of the component efficiencies, which are normally the turbine, drive system and generator

For a typical small hydro system the turbine efficiency would be 85%, drive efficiency 95% and generator efficiency 93%, so the overall system efficiency would be: $0.85 \times 0.95 \times 0.93 = 0.751$ i.e. 75.1%

See; <https://www.renewablesfirst.co.uk/hydropower/hydropower-learning-centre/how-much-power-could-i-generate-from-a-hydro-turbine/>

Determining the plant layout as per the safety involved

Seven ways to make a hydropower station a safer workplace (30 September 2015 by Ambrose Canning)

- Designing safety into hydropower stations
- Planning ahead to control risks
- Safety upgrades for older hydropower stations
- Station evacuation
- Flood protection
- Fire and smoke control
- Emergency and crisis management

SEE; <https://www.hydropower.org/blog/seven-ways-to-make-a-hydropower-station-a-safer-workplace>

Summary of standards (Safe Health Work 2016)

Safety and health organization

This section ascertain whether the company has safety departments, committees, and sectional safety committee.

Health and safety policy

The reason for this section is to determine whether the hydroelectric power plant station's safety management has Health and Safety Policies and whether these policies are implemented effectively as required by OSH standard.

Accident reporting, investigation, and analysis

This concerns accident reporting, investigation, and analysis as required by the Occupational Health and Safety Authority

Safety inspection

This captures data from the respondents concerning safety inspection.

Hazard identification and control

This covers company's hazard identification and control as required by Occupational Safety and Health Standard.

Safety operations and procedures

This concerns the company's safety operations and procedures.

Equipment safety

This section of the questionnaire gathered data on equipment safety at the power plant

15.3.3.3 Self-Assessment

1. State the safety requirement in a power station
2. Draw the schematic layouts of the following power stations
 - a) Geothermal power plant
 - b) Nuclear power plant
 - c) Solar power plant
 - d) Wind power plant
3. Write the equations of the power developed in the following types of power plant
 - a) Hydro Power plant
 - b) Wind power plant
4. Explain hazard identification and control
5. Perform calculations in tidal power station

15.3.3.4 Tools, Equipment, Supplies and Materials for the specific learning outcome

- Actual of power plants
- Models of electric power plant
- OSHA standards
- IEE regulations
- Charts

15.3.3.5. References

1. <https://www.hydropower.org/blog/seven-ways-to-make-a-hydropower-station-a-safer-workplace>
2. <https://openei.org/wiki/wind-energy>
3. Principles of power system by V.K Mehta
4. Quinlan M., Bohle P., Lamm F. 3rd ed. Palgrave Macmillan; South Yarra (Australia): 2010. Managing occupational health and safety: a multidisciplinary approach.
5. Dawson P., Zanko M. Tilde University Press; Prahran (Australia): 2011. Social innovation at work: sustainable OHS in HRM; pp.

6. Nichols T., Walters D., Tasiran A.C. Trade unions, industrial mediation and industrial safety. Evidence from the UK. J Ind Relat. 2007.
7. Environmental, Health, and Safety Approaches for Hydropower Projects - March 2018
8. Employee Safety: Safety at Hydro Plants - 05/01/2009

15.3.4. Learning Outcome No. 3. Demonstrate understanding in the operation of generation station

15.3.4.1 Learning Activities

Learning Outcome No. 3. Demonstrate understanding in the operation of generation station	
Learning Activities	Special Instructions
• Using model power station, visiting links, observing video tapes, visiting actual power station and outline the constituents of the generating station as per the type of generating station. • Using model power station, visit links, observing video tapes, visiting actual power station and establishing the operation of the constituent component in generating stations by their standard operating procedure. • Using model power station, visiting links, observing video tapes, visiting actual power station and outlining energy conversions from one form to another in line with the operation of the constituent components of the plant.	• Visit a power plant • Visit the various links • Observe model power stations • Observe video tapes • Visit different websites • Hold consultations and discussions

15.3.4.2 Information Sheet No. 15/ LO3

The constituents of the generating station as per the type of generating station

a) Hydroelectric power station

It constitutes

- The dam

It is the wall created which stores water and develops the water head

It is made of masonry stone and concrete or concrete alone

- Spill ways

It is an overflow used to discharge the surplus water from the storage reservoir into the river on the downstream side of the dam

- Head work:

It consists of the diversion structure at the head of an intake. They include racks for diverting floating debris, booms, sluices, sediments and valves.

- Surge Tank;

It is tank open at the top in which water level rises or falls to reduce the pressure swings in the penstock

- Pen stock:

It can be open or closed conduits which carry water to the turbines. They are made of steel or reinforced concrete.

- **Water Turbine:-**

Used to convert the energy of falling water into mechanical energy.

In a hydro power station they are either of

- Impulse type
- Reaction type

- **Alternator:**

It converts mechanical energy into electrical energy

- **Transformer:**

It is mainly for stepping up and stepping down voltage

- **Circuit breaker.**

It provides protection to the power plant equipment in case of excess current flowing in a certain circuit

b) Nuclear Power station

- **Heat exchanger**

This is where the steam is developed for driving the turbine

- **Nuclear reactor**

It is the components in which the nuclear fuel is subjected to nuclear fission. It controls the chain reaction that starts after the fission is done

- **Steam turbine:**

It converts the kinetic energy of the steam into mechanical energy which rotates the alternator.

- **Alternator:**

It converts the mechanical energy into electrical energy

c) Gas Turbine Power plant

- **Compressor:-**

A rotary type of compressor is used. It is used to increase the air pressure.

- **Regenerator:**

It is a device which recovers heat from the exhaust gases of the turbine. It consists of a nest of tubes contained in a shell

- **Combustion chamber.**

The heat is added to the air at high pressure by burning oil. This facilitates atomization of oil and its thorough mixing with air.

- **Gas turbine**

Converts the kinetic energy of the gas at high temperature and pressure into turning mechanical energy.

- **Alternator.**

It converts mechanical energy of the turbine into electrical energy.

- **Starting Motor:**

It is meant to start the compressor before starting the turbine.

- The Batteries

For energizing the motor.

d) Solar Power station

- PV module

It converts the solar energy into electrical energy

- Charge controller

It prevents the overcharging and over discharging of the battery.

- Battery

It stores the electrical energy

- Inverter:

It converts the D.C current / voltage into A.C current/voltage.

- Fuel/circuit breaker:

It protects the circuit in case of excess current in the circuit.

e) Diesel Power station

- Fuel supply system:

It has strainers, storage tank, fuel transfer and all day storage tank

- Air Intake System:

It supplies air to the engine for fuel combustion

It has pipes and filters

- Exhaust system:

It assets to remove the engine exhausts gas outside the building and discharge it into atmosphere.

It also incorporates a silencer.

- Cooling system

It provides the cooking in order to keep the temperate of the engine parts within the safe operating limits.

It has the cooling tower, water source and pumps.

- Lubricating system

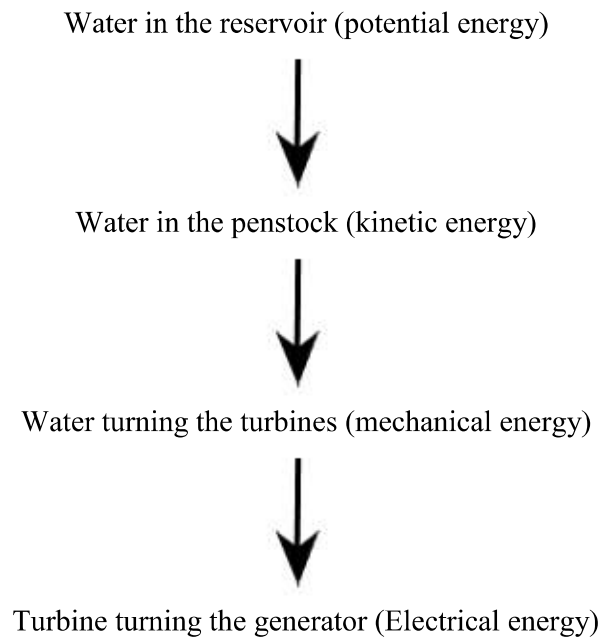
It minimizes the wear of rubbing surfaces of the engine. It consists of lubricating oil tank, the filers and oil cooler.

- Engine starting system:

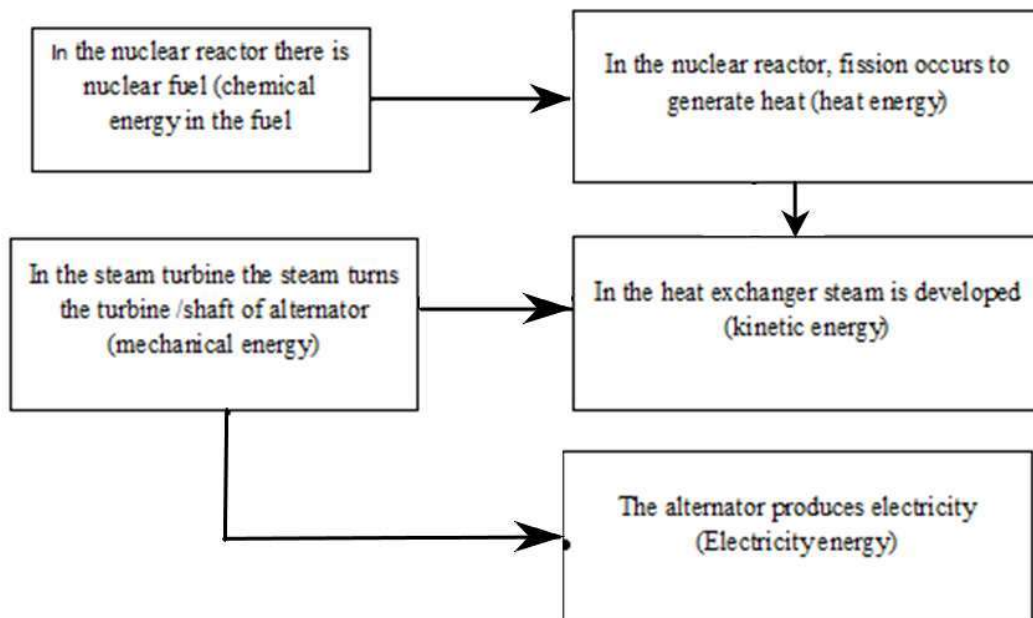
It is an arrangement to rotate the engine initially while starting until firing starts and then with its own power.

Determining Energy conversations from one form to another in line with the operation of the constituents of the plant.

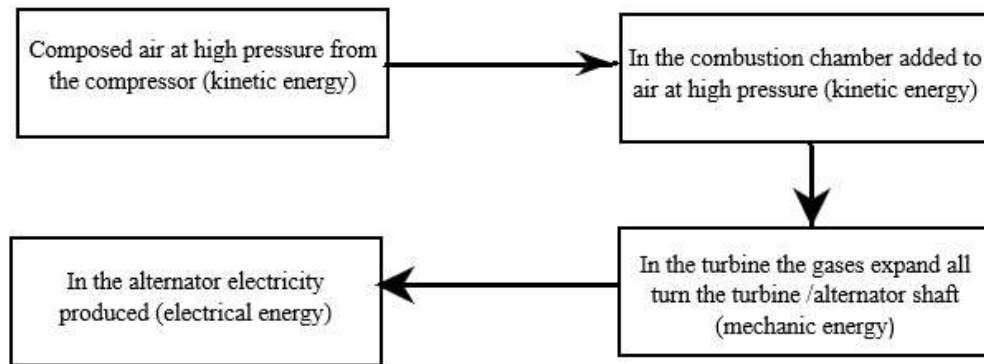
Hydroelectric Power Station



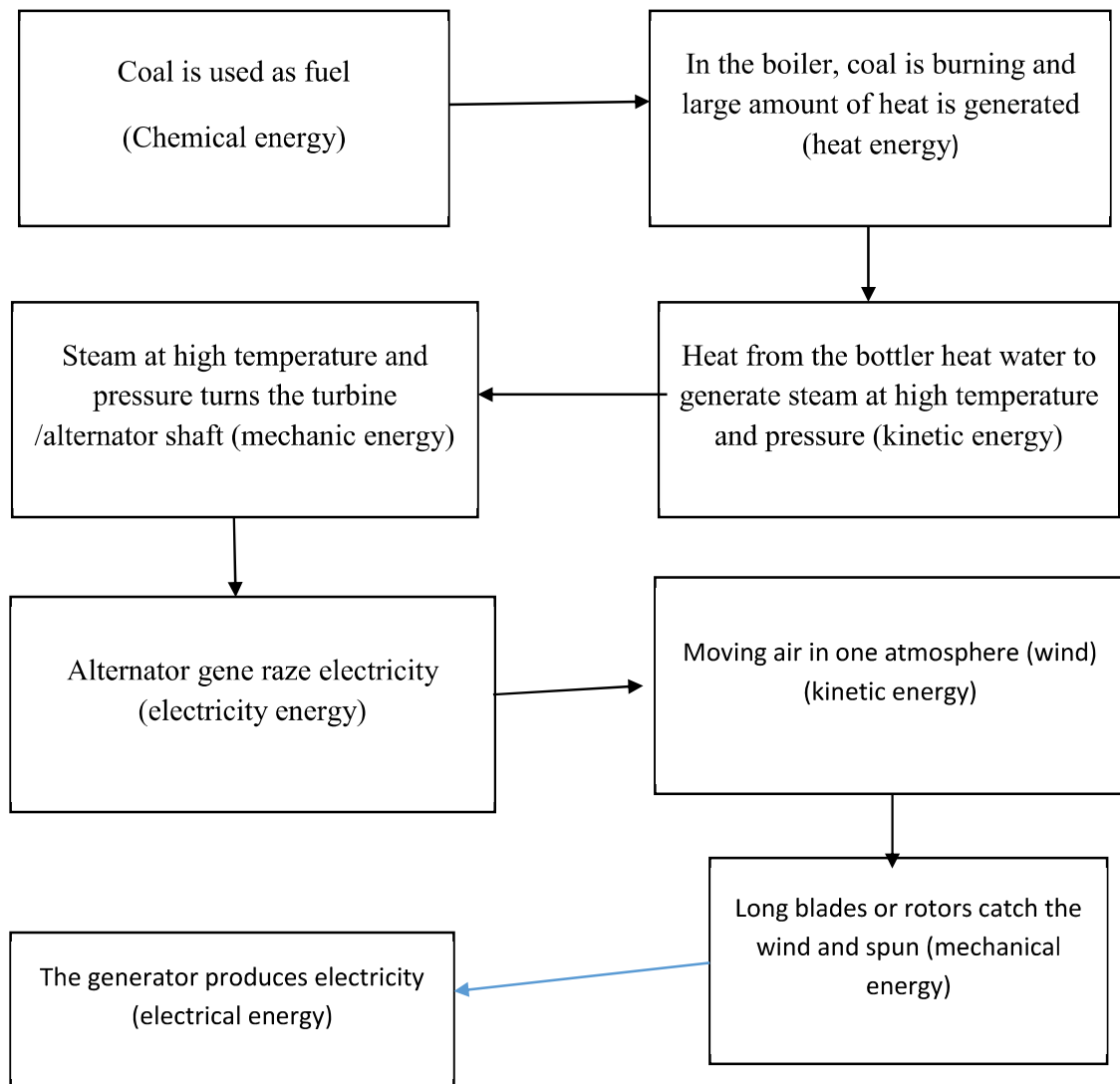
Nuclear Power Station



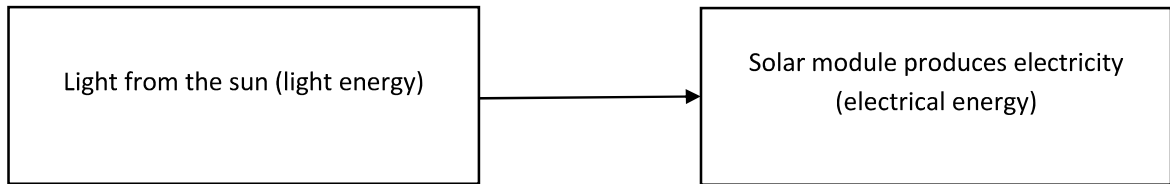
Gas Turbine Power Plant



Thermal power station



Solar power station



$$\text{Power} = (\text{Height of Dam}) \times (\text{River Flow}) \times (\text{Efficiency}) / 11.8$$

2.3.2.3 Self-Assessment

1. What is the advantage of sectionalizing of power plant?
 - a) High reliability
 - b) Low capital cost
 - c) Low maintenance
 - d) Easy operation
2. The area under the load curve represents _____
 - a) the average load on power system
 - b) maximum demand
 - c) number of units generated
 - d) load factor
3. Which of the following is equal to the maximum demand?
 - a) The ratio of area under curve to the total area of rectangle
 - b) The ratio of area under curve and number of hours
 - c) The peak of the load curve
 - d) The area under the curve
4. State a generating station
5. Outline the constituents of the generating station as per the type of generating station.
6. Explain the operation of the constituent component of a geothermal power station.
7. Describe the constituents of the generating station as per the type of generating station
8. Outline energy conversions from one form to another of the following power station.
 - a) Diesel power station
 - b) Nuclear power station
 - c) Solar power station

Practical questions

1. Determine the constituents of the generating station by the type of the generating station
2. Determine Operation of the constituent components in generating stations by their standard operating procedure.

3. Demonstrate the conversion of energy through simulation from one form to another in line with the operation of the constituent components of the plant.

15.3.3.4. Tools, Equipment, Supplies and Materials for the specific learning outcome

- Actual of power plants
- Models of electric power plant
- OSHA standards
- IEE regulations
- Charts
- Video tapes

15.3.3.5 References

1. <https://www.hydropower.org/blog/seven-ways-to-make-a-hydropower-station-a-safer-workplace>
2. <https://openei.org/wiki/wind-energ>
3. Principles of power system by V.K Mehta

15.3.5 Learning Outcome No. 4. Demonstrate understanding in the operating sequence of generating station

15.3.5.1 Learning Activities

Learning Outcome No. 4. Demonstrate understanding in the operating sequence of generating station	
Learning Activities	Special Instructions
• Determining the operating sequence of the generating plant by the energy flow from model power station, visit links, observing video tapes, visiting actual power station and • Using model power station, visit links, observing video tapes, visiting actual power station and determining the operating sequence of the generating plant by the type of generating station. • Using model power station, visit links, observing video tapes, visiting actual power station and determining the operating sequence by the type of components used in the generating station.	• visit a power plant • visit the various links • Observe model power stations • Observe video tapes • Visit different websites • Hold consultations and discussions

15.3.5.2 Information Sheet No. 15/ LO4

Demonstrate understanding in the operating sequence of generating station

- Determining the operating sequence of the generating plant by the energy flow.
- Determining the operating sequence of the generating plant by the type of generating station.
- Determine the operating sequence by the type of components used in the generating station.

- iv. Determining the operating sequence of the generating plant by the energy flow.

Energy flow in a wind power plant

Air moving in the atmosphere (Kinetic energy)

Wind turning the turbine/Generator shaft (mechanical energy)

Generator producing energy (Electrical engineering)

Wind energy is a form of solar energy. Wind turbines convert the kinetic energy in the wind into mechanical energy. Then a generator converts the mechanical power (energy) into electrical energy (electricity).

(<https://openei.org/wiki/wind-energy>)

Nuclear Power Station Layout

Hydro Power Plant

For the energy flow on hydro power station, Diesel power station and others see- principles of power system by V.K Mehta.

Determining the operating sequence of the generating plant by the type of generating station.

Hydro Electrical Power Station.

Hydro-electric power station is, located in hilly places where dams are built and accommodate large water reservoirs.

A water head is formed by making a dam across a river or lake. From the dam water is directed to the water turbine. The water turbine captures the energy in the falling water and changes the hydraulic energy into mechanical energy at the turbine shaft. The turbines turn the alternator and then the alternator converts mechanical energy into electrical energy. Hydro power stations assist in controlling floods, water for irrigation and water for drinking.

See:- For the other types of power stations see- principles of power stations by V. K. Mehta,

15.3.5.3. Self-Assessment

1. Out of the following which one is not an unconventional source of energy?
 - a) Tidal power
 - b) Geothermal energy
 - c) Nuclear energy
 - d) Wind power
2. Out of the following which one is not an unconventional source of energy?
 - a) Tidal power
 - b) Geothermal energy
 - c) Nuclear energy

- d) Wind power
- 3. What is Demand factor?
 - a) Ratio of connected load to maximum demand
 - b) Ratio of average load to connected load
 - c) Ratio of maximum demand to the connected load
 - d) Ratio of kilowatt hour consumed to 24 hour
- 4. Which of the following is not a type of Combined Plant?
 - a) Sodium- mercury-Potassium plant
 - b) Gas turbine-Steam turbine plant
 - c) Thermionic steam plant
 - d) Thermoelectric steam plant
- 5. Which of these is not a component of nuclear reactor?
 - a) reactor core
 - b) refractor
 - c) control rod
 - d) biological shield
- 6. Which of these is not a merit of nuclear power?
 - a) amount of fuel required is small
 - b) plant requires a huge amount of area
 - c) demand for coal and oil is reduced
 - d) most economical in large quantities
- 7. Wind turbines convert the kinetic energy in the wind into mechanical energy.
TRUE OR FALSE?
- 8. This section must be related with the Performance Criteria, Required Knowledge and Skills and the Range as stated in the Occupational Standards.
- 9. Determining the operating sequence of the generating plant by the energy flow.
- 10. Outline the operating sequence of
 - a) Hydro generating station.
 - b) Diesel power station

Practical Exercises

1. Determining the operating sequence of the generating plant by the energy flow from model power station, visit links, observing video tapes, visiting actual power station
2. Demonstrate the operating sequence of the generating plant by the type of generating station. Sources of power generating can be model power station, visit links, observing video tapes, visiting actual power station and
3. Determining the operating sequence by the type of components used in the generating station, using model power station, visit links, observing video tapes, visiting actual power station

14.3.6.4. Tools, Equipment, Supplies and Materials for the specific learning outcome

- Actual of power plants
- Models of electric power plant
- OSHA standards
- IEE regulations
- Charts
- Video tapes

15.3.5.5. References

1. <https://www.hydropower.org/blog/seven-ways-to-make-a-hydropower-station-a-safer-workplace>
2. <https://openei.org/wiki/wind-energy>
3. Principles of power system by V.K Mehta